

CMS *Internal*pp $\sqrt{s} = 5.02$ TeV (28.0 pb $^{-1}$)

Events

J/ ψ data $15.0 < p_T < 20.0$, $|y| < 1.6$ $0.020 < \text{ctau3D} < 0.025$ mm

status/cov = 0/3

—•— **Data**— **Fit**- - - **Signal**- - - **Background** $N_{\text{sig}} = 5545 \pm 77.4$ $N_{\text{bkg}} = 598.4 \pm 32.4$

SR entries = 5483

 $\mu = 3.0978 \pm 0.00053$ $\sigma_{\text{G1}} = 0.01902 \pm 0.00053$ $\sigma_{\text{CB1}} = 0.03939 \pm 0.0012$

1400

1200

1000

800

600

400

200

0

-200

-400

-600

-800

-1000

-1200

-1400

-1600

-1800

-2000

-2200

-2400

-2600

-2800

-3000

-3200

-3400

-3600

-3800

-4000

-4200

-4400

-4600

-4800

-5000

-5200

-5400

-5600

-5800

-6000

-6200

-6400

-6600

-6800

-7000

-7200

-7400

-7600

-7800

-8000

-8200

-8400

-8600

-8800

-9000

-9200

-9400

-9600

-9800

-10000

-10200

-10400

-10600

-10800

-11000

-11200

-11400

-11600

-11800

-12000

-12200

-12400

-12600

-12800

-13000

-13200

-13400

-13600

-13800

-14000

-14200

-14400

-14600

-14800

-15000

-15200

-15400

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-18600

-18800

-19000

-19200

-19400

-19600

-19800

-20000

-20200

-20400

-20600

-20800

-21000

-21200

-21400

-21600

-21800

-22000

-22200

-22400

-22600

-22800

-23000

-23200

-23400

-23600

-23800

-24000

-24200

-24400

-24600

-24800

-25000

-25200

-25400

-25600

-25800

-26000

-26200

-26400

-26600

-26800

-27000

-27200

-27400

-27600

-27800

-28000

-28200

-28400

-28600

-28800

-29000

-29200

-29400

-29600

-29800

-30000

-30200

-30400

-30600

-30800

-31000

-31200

-31400

-31600

-31800

-32000

-32200

-32400

-32600

-32800

-33000

-33200

-33400

-33600

-33800

-34000

-34200

-34400

-34600

-34800

-35000

-35200

-35400

-35600

-35800

-36000

-36200

-36400

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-36800

-37000

-37200

-37400

-37600

-37800

-38000

-38200

-38400

-38600

-38800

-39000

-39200

-39400

-39600

-39800

-40000

-40200

-40400

-40600

-40800

-41000

-41200

-41400

-41600

-41800

-42000

-42200

-42400

-42600

-42800

-43000

-43200

-43400

-43600

-43800

-44000

-44200

-44400

-44600

-44800

-45000

-45200

-45400

-45600

-45800

-46000

-46200

-46400

-46600

-46800

-47000

-47200

-47400

-47600

-47800

-48000

-48200

-48400

-48600

-48800

-49000

-49200

-49400

-49600

-49800

-50000

-50200

-50400

-50600

-50800

-51000

-51200

-51400

-51600

-51800

-52000

-52200

-52400

-52600

-52800

-53000

-53200

-53400

-53600

-53800

-54000

-54200

-54400

-54600

-54800

-55000

-55200

-55400

-55600

-55800

-56000

-56200

-56400

-56600

-56800

-57000

-57200

-57400

-57600

-57800

-58000

-58200

-58400

-58600

-58800

-59000

-59200

-59400

-59600

-59800

-60000

-60200

-60400

-60600

-60800

-61000

-61200

-61400

-61600